

Gauge Field Theories on a Lattice

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Received June 17, 1977

We begin a rigorous, nonperturbative investigation of quantum field theories with local internal symmetries. We discuss the lattice approximation of Yang–Mills fields and of fermion fields in the Euclidean setup and we verify physical positivity for the Schwinger functions of these approximations. This implies the existence of a positive self-adjoint transfer matrix. We then prove existence and analyticity of the infinite volume limit of strongly coupled Yang–Mills theories on the lattice and we verify Wilson's confinement bound. Finally we present a rigorous treatment of the Higgs mechanism in lattice gauge theories.

1. INTRODUCTION

Quantum field theories with (non-abelian) local internal symmetry groups appear to be the most promising models for elementary particles interacting weakly, electromagnetically, and strongly. While there is a large class of quantum field theory models, without local internal symmetries (and therefore not containing gauge fields), which we think we understand quite well both on a perturbative and on a nonperturbative level, there is much less that is *known* about the mathematical structure of gauge field theories. On the level of formal power series expansions, much work has been done to show that gauge theories are renormalizable; see [1, 20, 34].

In this paper we begin an analysis of gauge field theories on a nonperturbative level. The general strategy is to regularize the theory through the introduction of cutoffs and to study subsequently the limit as the cutoffs are removed. Unfortunately the cutoffs always destroy some of the symmetries of the models, which then have to be recovered in the limit. Also, some cutoffs destroy the positivity of the metric

* Supported in part by the National Science Foundation under Grant PHY 75-21212.

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‡ Supported in part by the National Science Foundation under Grant MPS 74-22844.

of the Hilbert space of physical states. This should be avoided, because it seems to be quite complicated to exhibit positivity as a property that holds in the no-cutoff limit only. A “good” cutoff therefore preserves positivity and most of the symmetries. In particular, it should preserve gauge invariance. The lattice approximation first introduced by Wilson [35] and by Polyakov has these properties.

In Section 2 we describe the lattice approximation for quantum field theories in the Euclidean formulation. Our definition of lattice gauge fields follows [35, 2], but we treat the fermions somewhat differently from [35, 18, 21]. This permits us to follow the general scheme of [25] to prove that the lattice approximations satisfies the physical positivity condition (i.e., positive metric for the Hilbert space of physical states) and to construct a transfer matrix $T = e^{-H}$ with $H = H^* \geq 0$. We also show how to apply rigorously the ansatz of Faddeev and Popov [8].

In Section 3 we show how to remove the volume cutoff in a strongly coupled pure Yang–Mills theory. Using a cluster expansion we prove that the infinite volume limit of the Schwinger functions of the theory exists and is analytic in the (inverse) coupling constant and that the theory has a nonvanishing mass gap.

In Section 4 we study the Higgs mechanism [5, 15, 16]. We discuss theories of gauge fields coupled to a matter field whose internal symmetries are spontaneously broken. The presence of a gauge field prevents the appearance of massless Goldstone particles and the gauge field “acquires a mass.” We exhibit this mechanism in lattice theories for the case of total symmetry breakdown with no massless particles left.

In Section 5 we make use of the results of Section 3 to prove Wilson’s confinement bound [35] for strongly coupled Yang–Mills theories.

In the final section we discuss some speculations about the continuum limit and look at a simplified Yang–Mills model which can be solved exactly.

We use some elementary notions and facts from the theory of Lie groups without comment. Standard references are, e.g., [17, 29].

Parts of this paper have already been reported in [24].

2. THE LATTICE APPROXIMATION

Euclidean Green’s functions of an arbitrary fermion-boson model uniquely define a relativistic quantum field theory if they satisfy the conditions of physical positivity, (anti)symmetry, Euclidean covariance, and a regularity condition [25].

In this section we introduce a lattice approximation for Euclidean quantum field theories, involving gauge fields, fermions, and bosons, and show that it satisfies the physical positivity condition (i.e., positive metric for the Hilbert space of physical states). This condition will continue to hold in the continuum limit, i.e., as we let the lattice spacing tend to zero. Following the general scheme introduced in [25] we then construct the Hamiltonian H , or the transfer matrix $T = e^{-H}$, in the language of statistical mechanics. It follows from the construction that $H = H^* \geq 0$.

For a definition of lattice boson fields we refer to [14]. Our definition of lattice gauge fields is standard, too [2, 35], whereas we treat the fermions somewhat differently

from [18, 21, 35]. In the present construction we try to preserve as much structure of the continuum theory as possible. In particular, chiral invariance is not destroyed by our lattice approximation.

We will always work with a hypercubical lattice $\mathcal{A}$ in $\mathbb{R}^d$ ($d \geq 2$) with lattice spacing a . For technical reasons (see Section 2.3) we choose $\mathcal{A}$ to be symmetric with respect to the coordinate axes with no points on these axes. A general point in $\mathcal{A}$ is thus of the form

$$x = (x^0, \dots, x^{d-1}), \quad x^i = a(\frac{1}{2} + n), \quad n \text{ integer}, \quad |x^i| < L_i.$$

The values of L_i determine the size of $\mathcal{A}$. We write $\mathcal{A}_\infty$ for the infinite lattice, $L_i = \infty$, $i = 1, \dots, d$. Furthermore we let e_μ be the vector of length a parallel to the μ th coordinate axis. (Ordered) pairs xy of nearest-neighbor lattice points $x, y \in \mathcal{A}$ are called (directed) *bonds*, closed loops $uvxy$ consisting of four bonds uv , vx , xy , and yu are called *plaquettes*. We will say that the bond xy (the plaquette $uvxy$) is in $\mathcal{A}$ if x and y (u, v, x , and y) are in $\mathcal{A}$.

For simplicity we assume the lattice spacing a to be 1 unless specified otherwise.

2.1. Lattice Yang-Mills Fields [2, 35]

Let $\mathcal{G}$ be a compact Lie group (the gauge group), let χ be a character on $\mathcal{G}$, and let $g_\cdot$ be a map from directed lattice bonds in $\mathcal{A}$ into $\mathcal{G}$ such that

$$g_{xy} = g_{yx}^{-1}.$$

Such a map $g_\cdot$, also denoted by $\{g_{xy}\}$, is called a “field configuration.” The space of all field configurations is denoted by $\mathcal{G}^{\mathcal{A}}$.

To each plaquette $P = uvxy$ in $\mathcal{A}$ there corresponds a conjugacy class $g_P \subset \mathcal{G}$ defined by

$$g_P = \text{conj class of } g_{uv}g_{vx}g_{xy}g_{yu}.$$

With every plaquette P we associate the expression

$$A_P = \frac{1}{2}(\chi(g_P) + \overline{\chi(g_P)}). \quad (2.1)$$

Notice that A_P is independent of the orientation and of the starting point of the path around P . More generally, with a closed path $C = x_1x_2 \cdots x_n$ on the lattice $\mathcal{A}$ (i.e., $x_1x_2, x_2x_3, \dots, x_{n-1}x_n$ and x_nx_1 are lattice bonds), we associate the expression $\chi(C) = \chi(g_{x_1x_2}g_{x_2x_3} \cdots g_{x_nx_1})$, where χ is again some character on $\mathcal{G}$.

The Yang-Mills action on the lattice $\mathcal{A}$ is defined by

$$A_{\mathcal{A}}^{\text{YM}} = \frac{1}{2g_0^2} \sum_{\mathcal{A}} A_P, \quad (2.2)$$

where g_0 is the Yang–Mills coupling constant and the sum $\sum_A$ runs over all plaquettes P in Λ .

With A_Λ^{YM} we define a Gibbs measure

$$d\mu_\Lambda = Z_\Lambda^{-1} e^{A_\Lambda^{\text{YM}}} dg_\infty,$$

$$Z_\Lambda = \int e^{A_\Lambda^{\text{YM}}} dg_\infty,$$

where dg_∞ is the infinite product $\prod dg_{xy}$ of Haar measures dg_{xy} , xy any bond in Λ_∞ . We define a *local observable* to be a function $\mathcal{F}$ in $L^\infty(dg_\infty)$ which depends only on *finitely many* bond variables g_{xy} . The expectation of $\mathcal{F}$ is defined by

$$\langle \mathcal{F} \rangle_\Lambda = \int \mathcal{F} d\mu_\Lambda. \quad (2.3)$$

A *local gauge transformation* is a map γ from lattice points $x \in \Lambda_\infty$ into the group $\mathcal{G}$ such that $\gamma_x = e$ (the identity element in $\mathcal{G}$) for all but finitely many points $x \in \Lambda_\infty$. Associated with γ there is a transformation of the bond variables

$$g_{xy} \rightarrow g_{xy}^\gamma \equiv \gamma_x g_{xy} \gamma_y^{-1}. \quad (2.4)$$

Both the A_Λ^{YM} and dg_∞ are invariant under local gauge transformations and so is the Gibbs measure $d\mu_\Lambda$. An observable $\mathcal{F}$ is *gauge invariant* if for all local gauge transformations γ we have $\mathcal{F} = \mathcal{F}^\gamma$, where $\mathcal{F}^\gamma$ is defined by

$$\mathcal{F}^\gamma(\{g_{xy}^\gamma\}) = \mathcal{F}(\{g_{xy}\}).$$

As in the continuum case we may use local gauge transformations to eliminate redundant degrees of freedom from the integral in (2.3).

Pick a bond rs in Λ . In order to eliminate the bond variable g_{rs} from the action we choose a local gauge transformation

$$\begin{aligned} \gamma_x &= e & \text{for } x \neq r, \\ &= g_{sr} & \text{for } x = r. \end{aligned}$$

Postponing integration over g_{rs} we transform the remaining variables in integral (2.3) as follows.

$$g_{xy} \rightarrow g_{xy}' \quad (xy \neq rs).$$

This eliminates g_{rs} from the action and from $\mathcal{F}$ if the latter is gauge invariant. Otherwise we introduce

$$\mathcal{F} = \int \mathcal{F}'(\{g_{xy}'\}, g_{rs}) dg_{rs}.$$

In both cases (2.3) can be replaced by an average with respect to the measure $Z_A^{-1} e^{A \gamma_A} \prod_{xy \neq rs} dg_{xy}$. In effect this procedure amounts to fixing the bond variable g_{rs} to e (for gauge-invariant $\mathcal{F}$). It is easy to show that one can simultaneously fix to e the variables of as many bonds as one can draw without ever closing a loop.

All this is just an example of a more general procedure which Faddeev and Popov have proposed for the continuum gauge theories [8]. There integration over all field configurations is replaced by integration over a "hypersurface" in $\mathcal{G}^A$ chosen such that every orbit $\{g_{xy}\}$ of the group of gauge transformations ($\{g_{xy}^\gamma\}$ fixed, $\{\gamma_x\}$ varying) intersects it at least once. While Faddeev and Popov had to argue purely heuristically, the lattice approximation offers itself for a rigorous *derivation* of the Faddeev-Popov ansatz. Let us give an example.

We pick a "time axis" parallel to one of the lattice generators, the e_0 -direction, say. Then we want to restrict integration in (2.3) to the "hypersurface" determined by

$$g_{x, x+e_0} = e, \quad \text{all } x. \quad (2.5)$$

(We call the gauge defined by (2.5) "radiation gauge.") Let us define a δ -function (as a distribution) on $\mathcal{G}$ by

$$\delta(g) \equiv \sum_{\sigma} d_{\sigma} \chi_{\sigma}(g). \quad (2.6)$$

Here $\{\chi_{\sigma}\}$ are the characters of a complete system of inequivalent d_{σ} -dimensional representations of the group $\mathcal{G}$. By the theorem of Peter and Weyl [17, 27, 29], definition (2.6) gives for continuous invariant functions f on $\mathcal{G}$

$$\int \delta(g) dg = 1 \quad \text{and} \quad \int f(g) \delta(g) dg = f(e).$$

Furthermore, for all field configurations $\{g_{xy}\}$

$$\int \prod_{x \in A} \delta(g_{x, x+e_0}^{\gamma}) d\gamma = 1. \quad (2.7)$$

Here $d\gamma$ is the product $\prod_{x \in A} d\gamma_x$ of Haar measures $d\gamma_x$. To prove (2.7) one makes repeated use of

$$\begin{aligned} \int \delta(g_{x-e_0, x}^{\gamma}) \delta(g_{x, x+e_0}^{\gamma}) d\gamma_x &= \sum_{\sigma, \sigma'} d_{\sigma} d_{\sigma'} \int \chi_{\sigma}(\gamma_{x-e_0} g_{x-e_0, x} \gamma_x^{-1}) \chi_{\sigma'}(\gamma_x g_{x, x+e_0} \gamma_{x+e_0}^{-1}) d\gamma_x \\ &= \sum_{\sigma} d_{\sigma} \chi_{\sigma}(\gamma_{x-e_0} g_{x-e_0, x} g_{x, x+e_0} \gamma_{x+e_0}^{-1}) \\ &= \delta(\gamma_{x-e_0} g_{x-e_0, x} g_{x, x+e_0} \gamma_{x+e_0}^{-1}). \end{aligned}$$

For gauge-invariant $\mathcal{F}$ we can now write

$$\begin{aligned}
 \int \mathcal{F} d\mu_{\Lambda} &= Z_{\Lambda}^{-1} \int \mathcal{F}(\{g_{xy}\}) \prod_{x \in \Lambda} \delta(g_{x, x+e_0}^{\gamma}) e^{A_{\Lambda}^{\text{YM}}} dg_{\infty} d\gamma \\
 &= Z_{\Lambda}^{-1} \int \mathcal{F}(\{g_{xy}\}) \prod_{x \in \Lambda} \delta(g_{x, x+e_0}) e^{A_{\Lambda}^{\text{YM}}} dg_{\infty} \\
 &= (Z_{\Lambda}^{\text{rad}})^{-1} \int \mathcal{F}^{\text{rad}}(\{g_{xy}\}) e^{A_{\Lambda}^{\text{YM}, \text{rad}}} dg_{\infty} \\
 &= \langle \mathcal{F}^{\text{rad}} \rangle_{\Lambda}^{\text{rad}},
 \end{aligned} \tag{2.8}$$

where on the second line we have made a change of variables $g_{xy} \rightarrow g_{xy}^{\gamma^{-1}}$ and used gauge invariance of $\mathcal{F}$ and of A_{Λ}^{YM} . $\mathcal{F}^{\text{rad}}$ and $A_{\Lambda}^{\text{YM}, \text{rad}}$ are obtained from $\mathcal{F}$ and A_{Λ}^{YM} by putting all the bond variables $g_{x, x+e_0}$ equal to e . Z_{Λ}^{rad} is defined in the obvious way and equals Z_{Λ} . Notice that due to (2.7) there are no Faddeev–Popov ghosts in the gauge defined by (2.5). This gauge will be useful in Section 3.3, where we prove physical positivity.

Finally let us define gauge averages. For $\mathcal{F}$ a function as above, we set

$$\bar{\mathcal{F}} = \int \mathcal{F}^{\gamma} \prod_x d\gamma_x, \tag{2.9}$$

where the product runs over all lattice points x in the support of $\mathcal{F}$. If we are already restricted to the radiation gauge, then only “time”-independent gauge transformations are allowed (i.e., $\gamma_x = \gamma_{x'}$ if the spatial components of x and x' agree), and the product in (2.9) runs over lattice points in a plane of constant x^0 only. Obviously we have

$$\langle \mathcal{F} \rangle_{\Lambda} = \langle \bar{\mathcal{F}} \rangle_{\Lambda} \tag{2.10}$$

and

$$\langle \mathcal{F} \rangle_{\Lambda}^{\text{rad}} = \langle \bar{\mathcal{F}} \rangle_{\Lambda}^{\text{rad}}. \tag{2.11}$$

2.2. Euclidean Fermi Fields on the Lattice

Free Euclidean fermion fields Ψ_{μ}^k were introduced in [26]. They were defined such that

$$\begin{aligned}
 \langle \Psi_{\alpha}^i(x) \Psi_{\beta}^k(y) \rangle &= \langle \bar{T} \psi_{\alpha}(iX_0, \mathbf{x}) \psi_{\beta}^{\dagger}(iy_0, \mathbf{y}) \rangle \\
 &= \frac{1}{(2\pi)^4} \int \left(\frac{m - i\gamma_{\mu}^E p_{\mu}}{m^2 + p^2} \right)_{\alpha\beta} e^{ip(x-y)} d^4p
 \end{aligned}$$

and

$$\langle \Psi_{\alpha}^i(x), \Psi_{\alpha}^k(y) \rangle = 0, \quad i, k = 1, 2; \alpha, \beta = 1, \dots, 4.$$

Here

$$\{\gamma_\mu^E, \gamma_\nu^E\} = 2\delta_{\mu\nu}, \quad \gamma_\mu^E p_\mu \equiv \sum_\mu \gamma_\mu^E p_\mu.$$

We may choose

$$\gamma_0^E = \gamma_0 = \begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix},$$

$$\gamma_k^E = i\gamma_k = i \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix}, \quad k = 1, 2, 3.$$

ψ and $\bar{\psi} = \psi^* \gamma_0$ are free relativistic fermion fields. (Notice that for simplicity we have set $d = 4$). To define a lattice approximation of this theory we introduce a Grassmann algebra $\mathcal{O}_A$, whose generators are $\Psi_\alpha^k(x)$, where $k = 1, 2$; $\alpha = 1, 2, 3, 4$, and $x \in A$. More precisely, with each lattice site $x \in A$ we associate a complex vector space $V_x = V_x^1 \oplus V_x^2$, where both V_x^i are s -dimensional ($s = \text{spinor degree of freedom}$). In each V_x^i we pick a basis denoted by $\Psi_\alpha^i(x)$, $\alpha = 1, \dots, s$. In fermion theories with internal symmetries, V_x^i is of the form $V_x^i = (V_x^i)_{\text{spin}} \otimes (V_x^i)_{\text{int}}$, where $(V_x^i)_{\text{int}}$ is the internal symmetry space which carries a d ($= \dim(V_x^i)_{\text{int}}$) dimensional unitary representation $U(\cdot)$ of the gauge group $\mathcal{G}$. A basis in V_x^i is now denoted by $\Psi_{\alpha a}^i(x)$, $a = 1, \dots, d$.

$\mathcal{O}_A$ is the Grassmann algebra over the vector space $V_A = \bigoplus_{x \in A} V_x$. Following Berezin [3] we define a linear functional on $\mathcal{O}_A$, written as an "integral" $\oint$ as follows. We set

$$\oint \bigwedge_{\alpha, a} (\Psi_{\alpha, a}^1(x) \wedge \Psi_{a, \alpha}^2(x)) = 1$$

and

$$\oint (\text{monomial in } \Psi_{\alpha, a}^k(x) \text{ of lower than maximal degree}) = 0.$$

Then we extend $\oint$ to all of $\mathcal{O}_A$ by linearity. The symbol $\wedge$ means exterior multiplication and will be dropped in the following.

Now we define the Euclidean fermion action by

$$A_A^F = \sum_{x \in A} [m \Psi^1(x) \Psi^2(x) - \frac{1}{2} \Psi^1(x) \gamma_\mu^E \{ \Psi^2(x - e_\mu) - \Psi^2(x + e_\mu) \}]. \quad (2.13)$$

This is a lattice version of the formal continuum expression

$$A_{\text{cont}}^F = \int \Psi^1(x) \{ m - \gamma_\mu^E \partial_\mu \} \Psi^2(x) d^4x,$$

which formally gives (2.10). Notice that in (2.13) the spinor indices have been suppressed; e.g., $\Psi^1(x) \Psi^2(x)$ means $\sum_\alpha \Psi_\alpha^1(x) \Psi_\alpha^2(x)$. Also, terms involving points

$x \pm e_\mu$ not contained in Λ are to be left out in the sum of (2.11). Notice that m can be any real number including zero.

If there are internal degrees of freedom, then the fermion field may be coupled to the gauge field. The action then becomes

$$A_A^F = \sum_{x \in \Lambda} [m \Psi^1(x) \Psi^2(x) - \frac{1}{2} \Psi^1(x) \gamma_\mu^E \{U(g_{x, x-e_\mu}) \Psi^2(x - e_\mu) - U(g_{x, x+e_\mu}) \Psi^2(x + e_\mu)\}]. \quad (2.14)$$

Again we have suppressed the spinor indices and the indices referring to internal degrees of freedom. For example, $\Psi^1(x) \gamma_\mu^E U(g_{xy}) \Psi^2(y)$ stands for

$$\sum_{\substack{\rho, \sigma \\ a, b}} \Psi_{\rho a}^1(x) (\gamma_\mu^E)_{\rho \sigma} U(g_{xy})_{ab} \Psi_{\sigma b}^2(y).$$

As observables $\mathcal{F}$ we take elements in the Grassmann algebra $\mathcal{O}_\Lambda$ whose coefficients may be (continuous) functions of the bond variables g_{xy} . The expectation of $\mathcal{F}$ is defined by

$$\begin{aligned} \langle \mathcal{F} \rangle_\Lambda &= Z_\Lambda^{-1} \int dg_\infty \oint (\mathcal{F} \exp(A_\Lambda^{\text{YM}} + A_\Lambda^F)) \\ Z_\Lambda &= \int dg_\infty \oint \exp(A_\Lambda^{\text{YM}} + A_\Lambda^F). \end{aligned} \quad (2.15)$$

A local gauge transformation now consists of (2.4) plus a substitution

$$\begin{aligned} \Psi_\alpha^1(x) &\rightarrow \Psi_\alpha^{1'}(x) = \bar{U}(h_x) \Psi_\alpha^1(x), \\ \Psi_\alpha^2(x) &\rightarrow \Psi_\alpha^{2'}(x) = U(h_x) \Psi_\alpha^2(x). \end{aligned} \quad (2.16)$$

The action $A = A_\Lambda^{\text{YM}} + A_\Lambda^F$ is invariant under (2.4) and (2.16). We leave it as an exercise to verify that the functional $\oint_\Lambda$, too, is invariant under (2.16); see [3].

2.3. Physical Positivity

In this section we show how to use the expectation $\langle \cdot \rangle$ to define a (positive metric) Hilbert space $\mathcal{H}$, the Hilbert space of physical states. Then there is a standard construction of the “transfer matrix” e^{-H} , where H is the Hamiltonian [25].

We call the e_0 -direction the “time direction.” The set $\mathcal{O}^+$ of functions at positive times is defined to be the Grassmann algebra generated by $\{\Psi^k(x), x^0 > 0\}$ with coefficients which are functions $f(\{g_{xy}\})$ of the bond variables g_{xy} , $x^0 > 0$, $y^0 > 0$. Correspondingly we define $\mathcal{O}^-$. An antilinear map $\Theta: \mathcal{O}^+ \rightarrow \mathcal{O}^-$ is defined as follows:

$$\Theta f(\{g_{x,y}\}) \Psi^{k_1}(x_1) \cdots \Psi^{k_n}(x_n) = \bar{f}(\{g_{\partial x}, \partial_y\}) (\gamma_0^E \Psi^{3-k_n})(\partial x_n) \cdots (\gamma_0^E \Psi^{3-k_1})(\partial x_1),$$

where $\partial x = (-x^0, x^1, x^2, x^3)$ and $k_i = 1$ or 2 .

The main result of this section is contained in the following theorem.

THEOREM 2.1. With $\langle \cdot \rangle$ defined as in (2.15) and $\mathcal{F}$ a gauge-invariant function in $\mathcal{O}^+$,

$$\langle \Theta \mathcal{F} \cdot \mathcal{F} \rangle \geq 0.$$

Proof. We pass to the radiation gauge where all the variables belonging to bonds parallel to the time direction are fixed to e . The algebras of observables are now denoted by $\mathcal{O}_{\text{rad}}^{(\pm)}$. We define a cone $\mathcal{P}$ of "positive" functions as the convex cone generated by functions of the form $(\Theta \mathcal{F}) \mathcal{F}$ with $\mathcal{F} \in \mathcal{O}_{\text{rad}}^+$.

LEMMA 2.2. If $A \in \mathcal{P}$, then

$$\int dg_{\alpha}^{\mathbf{A}} \int A \geq 0.$$

Proof of Lemma 2.2. It suffices to assume $A = (\Theta \mathcal{F}) \mathcal{F}$,

$$\mathcal{F} = \mathcal{F}_0(\{g_{xy}\}_{x^0 > 0}) \prod_{\substack{\alpha, a \\ x: x^0 > 0}} \Psi_{\alpha, a}^1(x) \Psi_{\alpha, a}^2(x).$$

But for such an $\mathcal{F}$

$$\int dg_{\alpha}^{\mathbf{A}} \int (\Theta \mathcal{F}) \mathcal{F} = \left| \int dg_{\alpha} \mathcal{F}_0 \right|^2 \geq 0.$$

LEMMA 2.3. $\mathcal{P}$ is a multiplicative cone; i.e., if A and B are in $\mathcal{P}$, then AB is in $\mathcal{P}$.

Proof of Lemma 2.3. Let $A = \sum (\Theta A_i) A_i$, $B = \sum (\Theta B_i) B_i$ with $A_i, B_i \in \mathcal{O}_{\text{rad}}^+$. We may assume that A_i and B_j are homogeneous elements of the Grassmann algebra of degrees a_i and b_j , resp. Then

$$\begin{aligned} AB &= \sum_{ij} (\Theta A_i)(\Theta B_j) A_i B_j (-1)^{a_i b_j} \\ &= \sum_{ij} (\Theta(B_j A_i)) A_i B_j (-1)^{a_i b_j} \\ &= \sum (\Theta(A_i B_j))(A_i B_j) \in \mathcal{P}. \end{aligned}$$

LEMMA 2.4. $\exp(A_A^{\text{YM, rad}} + A_A^{\text{F, rad}}) \in \mathcal{P}$.

Proof of Lemma 2.4. We write the exponent as a sum of three terms $A^+ + A^- + \Delta A$, such that $A^+ \in \mathcal{O}_{\text{rad}}^+$, $A^- = \Theta A^+ \in \mathcal{O}_{\text{rad}}^-$ and ΔA is the part of the exponent which couples positive and negative time variables. Obviously $e^{A^+ + A^-} = e^{A^+ + \Theta A^+} = (\Theta e^{A^+}) e^{A^+} \in \mathcal{P}$. It therefore suffices to show that $\Delta A \in \mathcal{P}$.

The part involving fermions is of the form

$$\begin{aligned} & \frac{1}{2} \sum_{\substack{x \in \mathcal{A} \\ x^0 = \frac{1}{2}}} [\Psi^2(x - e_0) \gamma_0^E \Psi^1(x) + \Psi^1(x - e_0) \gamma_0^E \Psi^2(x)] \\ &= \frac{1}{2} \sum_{\substack{x \in \mathcal{A} \\ x^0 = \frac{1}{2}}} [(\Theta \Psi^1(x)) \Psi^1(x) + (\Theta \Psi^2(x)) \Psi^2(x)] \in \mathcal{P}. \end{aligned}$$

The Yang–Mills part of ΔA is a sum of terms A_P^{rad} , where P is a plaquette which intersects the $x^0 = 0$ plane (see Fig. 1). We have (with $\chi(g) = \text{tr } D(g)$, D a unitary representation of $\mathcal{G}$)

$$\begin{aligned} A_P^{\text{rad}} &= \frac{1}{2} (\chi(g_{\partial y \partial x} g_{xy}) + \text{complex conjugate}) \\ &= \frac{1}{2} \left(\sum_{ij} D_{ij}(g_{\partial y \partial x}) D_{ji}(g_{xy}) + \text{c.c.} \right) \\ &= \frac{1}{2} \left(\sum_{ij} (\Theta D_{ji}(g_{xy})) D_{ji}(g_{xy}) + \text{c.c.} \right) \in \mathcal{P}. \end{aligned}$$

This proves Lemma 2.4.

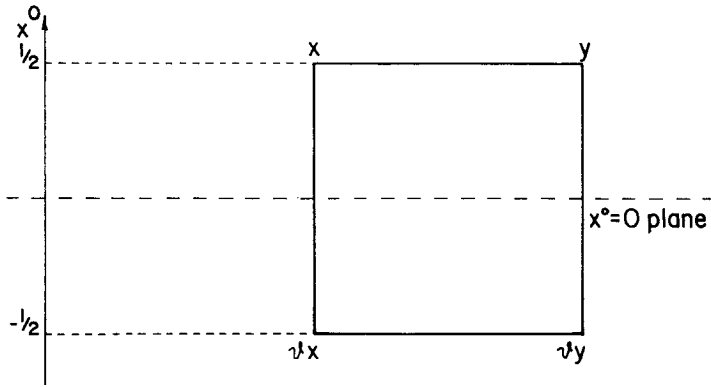


FIG. 1. The plaquette P .

Theorem 2.1 is an easy consequence of Lemmas 2.2–2.4.

We now define an expectation for $\mathcal{F} \in \mathcal{P}$ by

$$\langle \mathcal{F} \rangle_A^{\text{rad}} = Z_A^{-1} \int dg_\infty \int \mathcal{F} \exp(A_A^{\text{YM,rad}} + A_A^{\text{F,rad}}).$$

The physical Hilbert space $\mathcal{H}_A$ is defined to be the completion of $\mathcal{O}_{\text{rad}}^+ / \mathcal{N}$, with respect to the norm $\|A\|^2 = \langle (\Theta A) A \rangle_A^{\text{rad}}$, where

$$\mathcal{N} = \{A \in \mathfrak{U}_{\text{rad}}^+, \langle (\Theta A) A \rangle_A^{\text{rad}} = 0\}.$$

The natural map of $\mathcal{O}_{\text{rad}}^+$ into $\mathcal{H}_A$ will be denoted by W . Hence for $A, B \in \mathcal{O}_{\text{rad}}^+$

$$(WA, WB)_A = \langle (\Theta A) B \rangle_A^{\text{rad}},$$

where $(\cdot, \cdot)_A$ is the scalar product in $\mathcal{H}_A$. Notice that $\mathcal{H}_A$ contains a subspace $\mathcal{H}_{A,i}$ which is the image under W of the set of gauge-invariant functions $A \in \mathcal{O}^+$. At this point there is no reason to reject vectors in $\mathcal{H}_A$ but not in $\mathcal{H}_{A,i}$ (both $\mathcal{H}_A$ and $\mathcal{H}_{A,i}$ have of course a *positive metric*).

Let us sketch the construction of the transfer matrix and of the Hamiltonian. For simplicity we assume that we have already constructed the infinite volume limit $\Lambda \rightarrow \Lambda_\infty$ (see Section 3), and we drop the subscript Λ . Then we define T to be the operator on $\mathcal{O}_{\text{rad}}^+$ that shifts everything by two lattice units in time (e^0) direction. This determines an operator $T_{\mathcal{H}}$ on $\mathcal{H}$ by

$$(WA, T_{\mathcal{H}}WB) = \langle (\Theta A)(TB) \rangle^{\text{rad}}.$$

It is easy to show that $T_{\mathcal{H}}$ is a positive self-adjoint operator of norm less than 1. Hence we can write $T = e^{-2iH}$, where $H = H^* \geq 0$ is the Hamiltonian, e^{-H} is the transfer matrix. (Here we assume that T does not have zero as an eigenvalue. In the continuum limit this follows easily from the continuity of T^α as $\alpha \rightarrow 0$.)

Remarks. (1) Under a chiral transformation $\Psi_{\mu\alpha}$ transform as follows.

$$\begin{aligned}\Psi_{\alpha a}^1(x) &\rightarrow \sum_{\nu} (e^{i\alpha\gamma_5})_{\alpha\beta} \Psi_{\beta\alpha}^1(x), \\ \Psi_{\alpha a}^2(x) &\rightarrow \sum_{\nu} (e^{i\alpha\gamma_5})_{\beta\alpha} \Psi_{\beta\alpha}^2(x).\end{aligned}\tag{2.17}$$

The Euclidean fermion action (2.14) is invariant under (2.17), provided $m = 0$.

(2) Euclidean boson fields at time zero (i.e., $x^0 = 0$) can be naturally identified with the corresponding relativistic fields at time zero. In the fermion case one would like to identify $\Psi_\mu^1(0, \mathbf{x})$ and $\Psi_\mu^2(0, \mathbf{x})$ with the relativistic fields $\psi_\mu(0, \mathbf{x})$ and $\psi_\mu^\dagger(0, \mathbf{x})$, respectively. But this cannot be correct, because Ψ^1 and Ψ^2 always anticommute, while the relativistic fields do not. In the continuum case [26] this was not a real problem, because Euclidean fermi fields are too singular to exist at sharp times. This argument, however, does not work in a lattice theory. Our way out has been not to allow lattice points to be at $x^0 = 0$, but at $x^0 = \frac{1}{2} + \text{integer}$ only, thus avoiding the whole problem.

(3) If we consider a free fermion theory, then as we let the lattice spacing tend to zero the lattice approximation defined by (2.13) converges to the usual continuum theory in the sense that the lattice Schwinger functions converge to the continuum Schwinger functions as distributions. This, however, is not sufficient to give us the right continuum fermions if we couple the lattice fermions to some other field. Actually, perturbation theory suggests that the number of fermion degrees of freedom

gets multiplied by 2^d . To see this one should look at a closed fermion loop with lattice propagators

$$\langle \Psi_\alpha^1(x) \Psi_\beta^2(y) \rangle = \frac{1}{(2\pi)^4} \int_{|p_\mu| < \pi} d^4p e^{ip(x-y)} \left(\frac{m - i\gamma_\mu^E \sin p_\mu}{m^2 + \sum (\sin p_\mu)^2} \right)_{\alpha\beta}$$

and pass to the continuum limit. Because of the periodicity of the sine, one will end up with 2^d times the value for the same loop in a continuum theory. This problem of the lattice approximation has been noticed by various people and has been dealt with in different ways; see, e.g., [36]. Here we propose to simply add to A_A^F of (2.13) a term of the form

$$\Delta A_{A,1}^F = - \sum_{k \in A} \frac{1}{2} \Psi^1(x) (i\gamma_5) \sum_\mu \{ 2\Psi^2(x) - \Psi^2(x + e_\mu) - \Psi^2(x - e_\mu) \}. \quad (2.18)$$

Then $e^{A_{A,1}^F + \Delta A_A^F}$ is again of the form $(\Theta \mathcal{F}) \mathcal{F}$ with $\mathcal{F} \in \mathcal{O}_+^{\text{rad}}$ and hence physical positivity is still valid. To see this we proceed as in the proof of Lemma 2.4. Notice that for $x \in A$, $x^0 = -\frac{1}{2}$,

$$\frac{1}{2} \Psi^1(x) (\gamma_0^E + i\gamma_5) \Psi^2(x + e_0) = (\gamma_0^E \Psi^1)(x) \left(\frac{1 + i\gamma_0^E \gamma_5}{2} \right) \Psi^2(x + e_0)$$

is of the form $(\Theta A)A$ with $A \in \mathcal{O}_{\text{rad}}^+$ because $(1 + i\gamma_0^E \gamma_5)/2$ is a projection operator.

In the presence of gauge fields ΔA_A^F has to be modified appropriately and the argument goes through.

The Fourier transform of the propagator of the modified theory is

$$\frac{m - a^{-1}(i\gamma_\mu^E \sin p_\mu a + i\gamma_5 \sum (1 - \cos p_\mu a))}{m^2 + a^{-2}[\sum (\sin p_\mu a)^2 + (\sum (1 - \cos p_\mu a))^2]}, \quad (2.19)$$

where we have explicitly exhibited the dependence on the lattice spacing a . Again as $a \rightarrow 0$ this propagator converges to the continuum propagator but also the loops converge to the continuum loops. (The additional term in the denominator of (2.19) allows only the neighborhood of $p_\mu a = 0$ to contribute to the continuum limit). Hence the addition of (2.18) to the fermion action (2.13) prevents the creation of unwanted, additional fermion degrees of freedom. See [23, 31] for a discussion of convergence of approximate fermion theories.

3. STRONGLY COUPLED PURE YANG-MILLS THEORY: INFINITE VOLUME LIMIT, ANALYTICITY, AND MASS GAP

The construction of the infinite volume limit for a general lattice Yang-Mills theory is still an open problem. However, if the coupling is sufficiently strong, then there is a convergent “high-temperature” expansion which gives at the same time analyticity of the correlation functions in g_0^{-2} and the existence of a nonvanishing mass gap.

We first notice that in the particularly simple case of the Abelian gauge group $U(1)$, the existence of the infinite volume limit for arbitrary coupling strength follows from Griffith's inequalities which have been proved in sufficient generality by Ginibre [10]: The basic interaction term in this case is $A_P = \cos \varphi_P$. If we define P_+ to be the convex cone generated by

$$\{\cos n_{ij}\varphi_{ij} \mid n_{ij} \in \mathbb{Z}, i, j \in \mathbb{Z}^d\}$$

it follows from Ginibre's work [10] that for $\mathcal{F}_1, \mathcal{F}_2 \in P_+$

$$\begin{aligned} \langle \mathcal{F}_i \rangle_\Lambda &\geq 0, \\ \langle \mathcal{F}_1 \mathcal{F}_2 \rangle_\Lambda - \langle \mathcal{F}_1 \rangle_\Lambda \langle \mathcal{F}_2 \rangle_\Lambda &\geq 0. \end{aligned}$$

From this we conclude in the standard fashion that for $\mathcal{F} \in P_+$, $\langle \mathcal{F} \rangle$ increases monotonically in Λ to a limit $\langle \mathcal{F} \rangle_\infty$ and also that $\langle \mathcal{F} \rangle_\infty$ and $\langle \mathcal{F} \rangle_\Lambda$ decrease monotonically in g_0 (see also [6, 7]).

Let us now turn to the case of a pure Yang-Mills theory with arbitrary compact gauge group $\mathcal{G}$ and large coupling g_0 . We will use the fact that for large g_0

$$\exp((1/2g_0^2) A_P) \sim 1$$

to obtain a convergent "cluster expansion" for the expectation values of local observables (bounded functions of finitely many bond variables). In statistical mechanics, this would be called a high-temperature expansion. In quantum field theory it was first introduced by Glimm *et al.* [13] in their construction of $P(\varphi)_2$ models. Basically our method follows the arguments of [13], but we can simplify matters considerably due to the fact that

$$|A_P| \leq D,$$

where D is the dimension of the representation, of which χ is the character. Hence we expand

$$\begin{aligned} \exp \frac{1}{2g_0^2} \sum_{P \in \Lambda} (A_P + D) &= \prod_{P \in \Lambda} \left[1 + \left\{ \exp \frac{1}{2g_0^2} (A_P + D) - 1 \right\} \right] \\ &= \sum_{Q \subset \Lambda} \prod_{P \in Q} \rho_P. \end{aligned}$$

Here the sum runs over all subsets Q of plaquettes in Λ and

$$\rho_P \equiv \exp(1/2g_0^2)(A_P + D) - 1$$

fulfills

$$0 \leq \rho_P \leq \text{const}(1/g_0^2). \quad (3.1)$$

Now let $\mathcal{F}$ be a bounded (measurable) function of finitely many bond variables and let $Q_0 \equiv \text{supp } \mathcal{F}$ be the set of plaquettes which have bond variables occurring in $\mathcal{F}$. The finite volume expectation of $\mathcal{F}$ is given by

$$\begin{aligned} \langle \mathcal{F} \rangle_{\Lambda} &= Z_{\Lambda}^{-1} \int \mathcal{F} \exp \left(-\frac{1}{2g_0^2} \sum_{P \in \Lambda} (A_P + D) \right) dg_{\Lambda} \\ &= \sum_{Q \subset \Lambda} Z_{\Lambda}^{-1} \int \mathcal{F} \prod_{P \in Q} \rho_P dg_{\Lambda}, \end{aligned} \quad (3.2)$$

where

$$\begin{aligned} Z_{\Lambda} &= \int \exp \left(-\frac{1}{2g_0^2} \sum_{P \in \Lambda} (A_P + D) \right) dg_{\Lambda}, \\ dg_{\Lambda} &= \prod_{\Lambda} dg_{xy}, \end{aligned}$$

and the sum in (3.2) runs over all subsets Q of plaquettes in Λ . Next we split each Q into two disjoint sets $Q = Q_1 \cup Q_2$, where Q_1 is the union of all connected components of Q that have bonds in common with Q_0 ; $Q_2 = Q \setminus Q_1$. Then, obviously,

$$\langle \mathcal{F} \rangle_{\Lambda} = \sum_{Q \subset \Lambda} \left(\int \mathcal{F} \prod_{P \in Q_1} \rho_P dg_{\Lambda} \right) \left(\int \prod_{P \in Q_2} \rho_P dg_{\Lambda} \right) \cdot Z_{\Lambda}^{-1}.$$

We now fix Q_1 and sum over all Q_2 such that Q_2 has no bonds in common with either Q_0 or Q_1 . Observe that

$$\sum_{Q_2} \int \prod_{P \in Q_2} \rho_P dg_{\Lambda} = Z_{\Lambda \setminus \overline{Q_1 \cup Q_0}},$$

where $\overline{Q_1 \cup Q_0}$ is the set of all plaquettes which have bonds in common with plaquettes in $Q_1 \cup Q_0$. Therefore

$$\langle \mathcal{F} \rangle_{\Lambda} = \sum_{\substack{Q_1 \text{ conn } Q_0 \\ Q_1 \subset \Lambda}} \int \mathcal{F} \prod_{P \in Q_1} \rho_P dg_{\Lambda} \cdot \frac{Z_{\Lambda \setminus \overline{Q_1 \cup Q_0}}}{Z_{\Lambda}}. \quad (3.3)$$

$Q_1 \text{ conn } Q_0$, $Q_1 \subset \Lambda$ means that the sum runs over all $Q_1 \subset \Lambda$ with the property that each connected component of Q_1 has at least one bond in common with Q_0 ; (3.3) is the *cluster expansion*. We will show that for $|g_0|$ sufficiently large, it converges absolutely and uniformly in $|\Lambda|$, the size of the volume Λ (g_0 may be complex).

THEOREM 3.1. *Suppose $|g_0|$ is sufficiently large. Then there are constants $\alpha = \alpha(\mathcal{F}, d, \mathcal{G}, \chi)$ and $\beta = \beta(d, \mathcal{G}, \chi)$ such that*

$$\sum_{\substack{Q \subset \Lambda \\ Q \text{ conn } Q_0 \\ |Q| \geq K}} \left| \int \mathcal{F} \prod_{P \in Q} \rho_P dg_{\Lambda} \frac{Z_{\Lambda \setminus \overline{Q \cup Q_0}}}{Z_{\Lambda}} \right| \leq \alpha \left(\frac{\beta}{g_0^2} \right)^K. \quad (3.4)$$

Here $|Q|$ is the number of plaquettes in Q .

For the proof we will need three estimates.

LEMMA 3.2. For $|g_0|$ sufficiently large

$$2^{-|Q|} \leq |Z_{A \setminus Q}/Z_A| \leq 2^{|Q|}. \quad (3.5)$$

Remark. For real g_0 in fact

$$0 < Z_{A \setminus Q}/Z_A \leq 1.$$

This can be seen immediately by writing $Z_{A \setminus Q}$ and Z_A as integrals over all bond variables in Λ and observing that the integrand of $Z_{A \setminus Q}$ is smaller than the integrand of Z_A since

$$\exp(1/2g_0^2)(A_p + D) \geq 1.$$

LEMMA 3.3. There is a constant $c_1 = c_1(g, \chi)$ such that for all Q

$$\left| \int \mathcal{F} \prod_{P \in Q} \rho_P dg_\Lambda \right| \leq \|F\|_\infty \left(\frac{c_1}{|g_0|^2} \right)^{|Q|}.$$

LEMMA 3.4. Let $N(K)$ denote the number of possible sets Q of plaquettes in Λ such that $|Q| = K$ and such that all components of Q are connected with Q_0 . Then there are constants $c_2 = c_2(|Q_0|)$ and $c_3 = c_3(d)$ such that

$$N(K) \leq c_2 \cdot c_3^K.$$

The proof of Lemma 3.4 is a purely combinatorial estimate which can be derived as in [13]. Lemma 3.3 follows immediately from (3.1). We now give a proof of Lemma 3.2, which is essentially a simplified version of an argument given in [32]. It suffices to show that

$$\frac{1}{2} \leq |Z_{\Lambda \setminus \{P_0\}}/Z_\Lambda| \leq 2, \quad (3.6)$$

where $\{P_0\}$ consists of one plaquette $P_0 \subset \Lambda$. As above, we can expand

$$\begin{aligned} Z_\Lambda - Z_{\Lambda \setminus \{P_0\}} &= \sum_{\substack{Q \subset \Lambda \\ P_0 \in Q}} \int \prod_{P \in Q} \rho_P dg_\Lambda \\ &= \sum_{\substack{Q \subset \Lambda \\ Q \text{ conn } P_0}} \int \prod_{P \in Q} \rho_P dg_\Lambda Z_{\Lambda \setminus \overline{Q \cup \{P_0\}}}. \end{aligned} \quad (3.7)$$

Proceeding by induction we assume that (3.6) has been proved for $|\Lambda| = N - 1$,

and consider now the case $|A| = N$. From (3.7) and the induction hypothesis we obtain

$$\begin{aligned} \left| 1 - \frac{Z_A}{Z_{A \setminus \{P_0\}}} \right| &\leq \sum_{\substack{Q \subset A \\ Q \text{ conn } \{P_0\}}} \left| \int \prod_{P \in Q} \rho_P dg_A \right| \frac{Z_{A \setminus \overline{Q \cup P_0}}}{Z_{A \setminus \{P_0\}}} \\ &\leq \sum_{\substack{Q \subset A \\ Q \text{ conn } \{P_0\}}} \left(\frac{c_1}{|g_0|^2} \right)^{|Q|} \cdot 2^{|\overline{Q \cup P_0}|+1}, \quad \text{by Lemma 3.3,} \\ &\leq \sum_{K=1}^{\infty} \left(\frac{c_1}{|g_0|^2} \right)^K c_2 \cdot (\text{const})^K, \quad \text{by Lemma 3.4,} \end{aligned}$$

which can be made less than $\frac{1}{2}$ by choosing $|g_0|$ sufficiently large. Hence

$$|1 - (Z_A/Z_{A \setminus \{P_0\}})| < \frac{1}{2}$$

and (3.6) follows.

Proof of Theorem 3.1. This is an easy consequence of Lemmas 3.2 and 3.3. The left-hand side of (3.4) is bounded by

$$\sum_{k \geq K} N(k) \left| \frac{c_1}{g_0^2} \right|^k \| \mathcal{F} \|_{\infty} \leq c_2 \| \mathcal{F} \|_{\infty} \left(1 - \left| \frac{c_1 c_3}{g_0^2} \right| \right)^{-1} \left| \frac{c_1 c_3}{g_0^2} \right|^K.$$

A standard consequence of convergence of the cluster expansion is the existence of a mass gap (exponential clustering). We state the result and refer for a proof to [13]; see also [24].

THEOREM 3.5. *Let $\mathcal{F}_1, \mathcal{F}_2$ be functions of finitely many bond variables. Suppose g_0 is sufficiently large. Then there are constants $c = c(\mathcal{F}_1, \mathcal{F}_2)$ and $m > 0$ such that uniformly in A , for any lattice vector a ,*

$$|\langle \mathcal{F}_1 \mathcal{F}_2^a \rangle - \langle \mathcal{F}_1 \rangle \langle \mathcal{F}_2 \rangle| \leq c e^{-m|a|}. \quad (3.8)$$

Here $\mathcal{F}_2^a$ is the translate of $\mathcal{F}_2$ by the vector a .

Remark. It can be shown that

$$m \geq 4(\ln g_0^2 - \ln \beta), \quad (3.9)$$

with β as in Theorem 3.1. The reason is that in the cluster expansion for the left-hand side of (3.8), Q must contain at least $4(|a| - r)$ plaquettes; otherwise the corresponding term vanishes. (r is a constant depending on the size of the supports of $\mathcal{F}_1$ and $\mathcal{F}_2$, respectively.) The bound (3.9) is independent of the group $\mathcal{G}$ and in agreement with the results derived in a simplified model (Section 6).

Existence of the infinite volume limit and analyticity in $1/g_0^2$ of the infinite volume expectations are two more standard applications of the cluster expansion (see [13]). We just summarize the results.

THEOREM 3.6. *For g_0 sufficiently large, the infinite volume limit of the lattice Yang–Mills theories exists, has a mass gap $m \geq 4(\ln g_0^2 - \ln \beta)$, satisfies the condition of physical positivity, and is invariant under lattice translations.*

THEOREM 3.7. *The infinite volume expectations of a lattice Yang–Mills theory are analytic in $z = 1/g_0^2$ for $|z|$ sufficiently small.*

Remark. Translation invariance follows from uniqueness of the infinite volume limit.

4. THE HIGGS MECHANISM

In this section we discuss theories in which the gauge field is coupled to a “matter” field whose internal symmetry is spontaneously broken. Due to the presence of the gauge field, no massless Goldstone particles appear, but the gauge field “acquires a mass.” This is called the Higgs mechanism [5, 15, 16]. In field theory it is usually discussed on a semiclassical level. For a discussion on the axiomatic level see [33]. Here we prove the occurrence of the effect for lattice theories, at least for the case of total symmetry breakdown, where there are no massless degrees of freedom left.

In order to unify the treatment of gauge field coupling to matter fields with complete symmetry breakdown, we introduce a matter field as follows: With each lattice site x we associate a pair $(R(x), g(x))$, where $R(x)$ takes values in $\mathbb{R}_+ = [0, \infty)$ and $g(x)$ is an element in the gauge group $\mathcal{G}$ (polar coordinates!). Let us look at two examples:

(1) $\mathcal{G} = U(1)$. In this case the matter field is actually a complex scalar field $\Phi(x)$ with $R(x) = |\Phi(x)|$ and $g(x) = \Phi(x)/|\Phi(x)|$; i.e., (R, g) is just the polar coordinate representation of $\Phi(x) = R(x) e^{i\vartheta(x)}$, $g(x) = e^{i\vartheta(x)}$.

(2) $\mathcal{G} = SU(2)$. Now the matter field is a two-dimensional complex vector field

$$\Phi(x) = \begin{pmatrix} \Phi_1(x) \\ \Phi_2(x) \end{pmatrix}$$

with $R(x) = |\Phi(x)| = (|\Phi_1(x)|^2 + |\Phi_2(x)|^2)^{1/2}$ and $g(x)$ is uniquely defined by

$$\Phi(x) = \begin{pmatrix} \Phi_1(x) \\ \Phi_2(x) \end{pmatrix} = g(x) \begin{pmatrix} R(x) \\ 0 \end{pmatrix}.$$

We now define the action for the coupled system in a finite volume Λ .

$$A_\Lambda = \frac{1}{2g_0^2} \sum_{p \in \Lambda} (A_p - D) + \lambda \sum_{xy \in \Lambda} \chi(g(x)^{-1} g_{xy} g(y)) R(x) R(y) - \sum_{x \in \Lambda} (V(R(x)) + \frac{1}{2} d(x) \lambda D R(x)^2). \quad (4.1)$$

Here the first term is the pure Yang-Mills contribution $A_p = \chi(g_p)$ as in (2.1) (for simplicity we write here and in the following $\chi(g)$ instead of $\text{Re } \chi(g)$), $D = \chi(e)$; the second term gives the coupling between matter field and gauge field, while V is a function on $\mathbb{R}_+$ with a sharp absolute minimum at $R = \bar{R} \neq 0$. More precisely, the classes of potentials we will be considering are as follows:

$v(\bar{R}, c_1, c_2)$ = set of Lebesgue measurable functions V on $\mathbb{R}_+$, satisfying

$$\begin{aligned} V(R) &\geq c_1(R - \bar{R})^2 & \text{for all } R \in \mathbb{R}_+, \\ V(R) &\leq c_2(R - \bar{R})^2 & \text{for } |R - \bar{R}| \leq \bar{R}/2. \end{aligned} \quad (4.2)$$

Notice that $V(R) = \gamma(R^2 - \bar{R}^2)^2$ is in $v(\bar{R}, \gamma\bar{R}^2, \gamma(\frac{5}{2}\bar{R}^2))$ for any choice of $\bar{R}$ and of γ .

The term $\frac{1}{2} d(x) \lambda D(R(x))^2$ in (4.1) is added for convenience, as will be seen later. $d(x)$ is the number of bonds in Λ with one endpoint at x , i.e., for x not on the boundary $\partial\Lambda$ of Λ , $d(x)$ is simply twice the dimension d of space time. (See formulas (4.9)–(4.11).

The Gibbs measure is now defined by

$$d\mu_\Lambda = Z_\Lambda^{-1} e^{A_\Lambda} dv_\Lambda, \quad (4.3)$$

where

$$Z_\Lambda = \int e^{A_\Lambda} dv_\Lambda$$

and

$$dv_\Lambda = \prod_{xy \in \Lambda} dg_{xy} \prod_{x \in \Lambda} dg(x) \prod_{x \in \Lambda} R(x)^k dR(x) \quad (4.4)$$

($dR(x)$ is Lebesgue measure on $\mathbb{R}_+$, k is a positive integer depending on the gauge group $\mathcal{G}$; dg_{xy} and $dg(x)$ denote again Haar measure on $\mathcal{G}$).

In the case of the examples looked at before, we can bring (4.3) into a more familiar form.

(1) $\mathcal{G} = U(1)$. We choose $k = 1$, $g_{xy} = e^{i\vartheta_{xy}}$, $A_p = \cos \vartheta_p$. Then (4.1) is the same as

$$\begin{aligned} A_\Lambda &= \frac{1}{2g_0^2} \sum_{p \in \Lambda} (\cos \vartheta_p - 1) + \lambda \text{Re} \sum_{xy \in \Lambda} \Phi^*(x) \cos \vartheta_{xy} \Phi(y) \\ &\quad - \sum_{x \in \Lambda} (V(|\Phi(x)|) + d\lambda |\Phi(x)|^2), \\ dv_\Lambda &= \prod_{xy \in \Lambda} \frac{d\vartheta_{xy}}{2\pi} \prod_{x \in \Lambda} d(\text{Re } \Phi(x)) \cdot d(\text{Im } \Phi(x)). \end{aligned}$$

(2) $\mathcal{G} = SU(2)$. Choose $k = 3$, $A_P = \text{tr } g_P$. Then

$$\begin{aligned} A_\Lambda &= \frac{1}{2g_0} \sum_{P \in \Lambda} (\text{Tr } g_P - 2) + \lambda \sum_{xy \in \Lambda} 2 \text{Re}(\Phi(x), g_{xy} \Phi(y)) \\ &\quad - \sum_{x \in \Lambda} (V(|\Phi(x)|) + d\lambda |\Phi(x)|^2), \\ dv_\Lambda &= \prod_{xy \in \Lambda} dg_{xy} \prod_{\substack{x \in \Lambda \\ j=1,2}} d(\text{Re } \Phi_j(x)) d(\text{Im } \Phi_j(x)). \end{aligned}$$

The measure (4.3) is invariant under joint gauge transformations of gauge field and matter field

$$g(x) \mapsto g^\gamma(x) = \gamma_x g(x), \quad g_{xy} \mapsto g_{xy}^\gamma = \gamma_x g_{xy} \gamma_y^{-1}. \quad (4.5)$$

We now use this gauge invariance to eliminate redundant degrees of freedom. For each bond xy we define new bond variables

$$h_{xy} = g^{-1}(x) g_{xy} g(y), \quad (4.6)$$

and we denote by h_P the product of the four new bond variables belonging to the plaquette P (of course $h_P = g_P$!). We again write $A_P = \chi(h_P)$. Then, in terms of the new bond variables, A_Λ in (4.1) becomes

$$\begin{aligned} \hat{A}_\Lambda &= \frac{1}{2g_0^2} \sum_{P \in \Lambda} (A_P - D) + \lambda \sum_{xy \in \Lambda} \chi(h_{xy}) R(x) R(y) \\ &\quad - \sum_{x \in \Lambda} (V(R(x)) + \tfrac{1}{2}d(x) \cdot \lambda D R(x)^2) \end{aligned} \quad (4.7)$$

and

$$\begin{aligned} dv_\Lambda &= \prod_{xy \in \Lambda} dh_{xy} \prod_{x \in \Lambda} dg(x) \prod_{x \in \Lambda} R(x)^k dR(x) \\ &= d\hat{\nu}_\Lambda \cdot \prod_{x \in \Lambda} dg(x). \end{aligned}$$

The measure $d\mu_\Lambda$ (4.3) factorizes

$$d\mu_\Lambda = d\hat{\mu}_\Lambda \cdot \prod_{x \in \Lambda} dg(x)$$

where

$$d\hat{\mu}_\Lambda = \hat{Z}_\Lambda^{-1} e^{\hat{A}_\Lambda} d\hat{\nu}_\Lambda$$

and

$$\hat{Z}_A = \int e^{\hat{A}_A} d\hat{\nu}_A = Z_A.$$

Let $\mathcal{F}$ be a local observable (a bounded measurable function of the field variables g_{xy} , g_x and $R(x)$ with $x, y \dots$ contained in a subregion of A). Then

$$\langle \mathcal{F} \rangle_A = \int \mathcal{F} d\hat{\mu}_A,$$

where $\mathcal{F}_A$ is the "gauge average" of $\mathcal{F}$ defined by

$$\mathcal{F}_A(\{h_{xy}\}, \{R(x)\}) \equiv \int \mathcal{F}(\{g_{xy}^\gamma\}, \{g^\gamma(x)\}, \{R(x)\}) \prod_{x \in A} d\gamma_x. \quad (4.8)$$

It is therefore sufficient to study the reduced measure $d\hat{\mu}_A$ and gauge-invariant observables. From now on we will drop the hat on the quantities. As in the previous chapter we construct the infinite volume limit using a convergent cluster expansion. We begin by splitting A_A in a suitable way into a local part and a coupling part. We decompose

$$A_A = A_A^{(0)} + A_A^{(1)},$$

with

$$A_A^{(0)} = \lambda \sum_{xy \in A} (\chi(h_{xy}) - D) \bar{R}^2 - \sum_{x \in A} V(R(x)), \quad (4.10)$$

$$\begin{aligned} A_A^{(1)} = & \frac{1}{2g_0^2} \sum_{p \in A} (A_p - D) + \lambda \sum_{xy \in A} (\chi(h_{xy}) - D)(R(x)R(y) - \bar{R}^2) \\ & - \frac{1}{2}\lambda D \sum_{xy \in A} (R(x) - R(y))^2. \end{aligned} \quad (4.11)$$

According to the splitting (4.9) of A_A we write the measure $d\mu_A$ as

$$d\mu_A = Z_A^{-1} e^{A_A^{(1)}} d\mu_A^{(0)}, \quad Z_A = \int e^{A_A^{(1)}} d\mu_A^{(0)}, \quad (4.12)$$

where

$$d\mu_A^{(0)} = (Z_A^{(0)})^{-1} e^{A_A^{(0)}} d\nu, \quad Z_A^{(0)} = \int e^{A_A^{(0)}} d\nu_A. \quad (4.13)$$

The measure $d\mu_A^{(0)}$ factorizes

$$d\mu_A^{(0)} = (Z_A^{(0)})^{-1} \prod_{xy \in A} e^{\lambda(\chi(h_{xy}) - D)\bar{R}^2} dh_{xy} \cdot \prod_{x \in A} e^{-V(R(x))} R(x)^k dR(x). \quad (4.14)$$

The factor $e^{A^{(1)}}$ can be written in the form

$$e^{A^{(1)}} = \prod_{P \in \Lambda} (1 + \rho_P), \quad (4.15)$$

where

$$\begin{aligned} \rho_P &= e^{S_P} - 1, \\ S_P &= \frac{1}{2g_0^2} (\chi(h_P) - D) + \frac{\lambda}{2(d-1)} \\ &\quad \times \sum_{xy \in P} \left\{ (\chi(h_{xy}) - D)(R(x)R(y) - \bar{R}^2) - \frac{D}{2} (R(x) - R(y))^2 \right\}. \end{aligned} \quad (4.16)$$

Here $\sum_{xy \in P}$ runs over all four bonds xy of the plaquette P . The factor $1/2(d-1)$ in front of the sum accounts for the fact that any bond occurs in $2(d-1)$ different plaquettes. For plaquettes containing edges of the boundary of Λ that factor has to be modified.

Now we can repeat the arguments of Section 3 and write down a cluster expansion for the expectation of a local observable $\mathcal{F}$ with support Q_0 (see (3.3)).

$$\langle \mathcal{F} \rangle_\Lambda = \sum_{\substack{Q \text{ conn} \\ Q \subset \Lambda}} \int \mathcal{F} \prod_{P \in Q} \rho_P d\mu_\Lambda^{(0)} \cdot \frac{Z_{\Lambda \setminus Q \cup Q_0}}{Z_\Lambda}. \quad (4.17)$$

Our goal is to show that for any choice of g_0 and λ there is a large class of functions V for which the cluster expansion converges uniformly in Λ .

The main result of this section is the following theorem.

THEOREM 4.1. *For arbitrary positive values of the coupling parameters g_0 and λ , there are positive numbers $\bar{R}$, c_1 , c_2 such that for $V \in \mathfrak{v}(\bar{R}, c_1, c_2)$ the cluster expansion (4.17) converges uniformly in Λ , and the expectations of local observables have an infinite volume limit which has the physical positivity property, the exponential cluster property ($\Rightarrow$ mass gap) and is invariant under lattice translations.*

The proof of Theorem 4.1 follows step by step the arguments of Section 3 with one exception, the proof of Lemma 3.3. We restate it here in the appropriate form.

LEMMA 4.2. *For any positive g_0 and λ and for any $\epsilon > 0$, there are $\bar{R}$, c_1 , c_2 such that for $V \in \mathfrak{v}(\bar{R}, c_1, c_2)$*

$$\left| \int \mathcal{F} \prod_{P \in Q} \rho_P d\mu_\Lambda^{(0)} \right| \leq \| \mathcal{F} \|_\infty \cdot \epsilon^{|Q|}.$$

Remark. For simplicity we assume $\mathcal{F}$ to be bounded, such that $\| \mathcal{F} \|_\infty$ does not depend on $d\mu_\Lambda^{(0)}$.

Let us first try to understand qualitatively why Lemma 4.2 should be true. In

Section 3 we chose g_0 large and ρ_P was *uniformly* small. In the present situation, ρ_P may become large, but the values of h_P and $R(x)$ for which this happens will be strongly suppressed by the measure $d\mu_A^{(0)}$. Namely, given λ and g_0 , we choose V such that it has a sufficiently sharp minimum at sufficiently large $\bar{R}$. Then (4.14) tells us that the measure $d\mu_A^{(0)}$ strongly favors values of $R(x)$ which are very close to $\bar{R}$ and values of h_{xy} which are very close to the identity element. But these are exactly the values of the variables for which ρ_P is small.

Proof. Notice that there are at most $2(d-1)$ plaquettes containing a given bond and at most $r = 2d(d-1)$ plaquettes containing a given site. Hence any variable h_{xy} or $R(x)$ appears in at most r factors ρ_P . By repeated application of Hölder's inequality we therefore find

$$\left| \int \mathcal{F} \prod_{P \in Q} \rho_P d\mu_A^{(0)} \right| \leq \| \mathcal{F} \|_\infty \prod_{P \in Q} \| \rho_P \|_r$$

and Lemma 4.2 follows if we can show that under the stated conditions

$$\| \rho_P \|_r \leq \epsilon. \quad (4.18)$$

($\| \cdot \|_r$ is the norm in $L_r(d\mu_A^{(0)})$.) Writing ρ_P as $e^{S_P} - 1 = \int_0^1 S_P e^{tS_P} dt$ (see (4.16)), we find

$$\begin{aligned} \| \rho_P \|_r &= \left\| \int_0^1 S_P e^{tS_P} dt \right\|_r \\ &\leq \sup_{0 \leq t \leq 1} \| S_P e^{tS_P} \|_r \\ &\leq \| S_P \|_{2r} \cdot \sup_{0 \leq t \leq 1} \| e^{tS_P} \|_{2r}. \end{aligned} \quad (4.19)$$

We will get separate estimates of the two factors on the right-hand side of (4.19).

LEMMA 4.3. *Suppose $V \in \mathfrak{v}(\bar{R}, c_1, c_2)$ and the range of c_1 and c_2 is restricted by*

$$c_1 > [1 + (32dD) \lambda \bar{R}]^2, \quad 1 < c_2/c_1 < \alpha \text{ (fixed)}. \quad (4.20)$$

Then there is a constant c , depending on α and k only such that uniformly in $\bar{R} > 1$,

$$\sup_{0 \leq t \leq 1} \| e^{tS_P} \|_{2r} \leq c. \quad (4.21)$$

Furthermore, given $\epsilon > 0$ there is an $\bar{R}_0$ such that for $\bar{R} \geq \bar{R}_0$ and c_1, c_2 restricted as above,

$$\| S_P \|_{2r} \leq c^{-1} \epsilon. \quad (4.22)$$

Clearly Lemma 4.3 implies (4.18) and thus Lemma 4.2.

Notice that by (4.16)

$$\begin{aligned} S_P &\leq \frac{\lambda D}{(d-1)} \sum_{xy \in P} |R(x) R(y) - \bar{R}^2| \\ &\leq \frac{\lambda D}{d-1} \sum_{x \in P} \{2\bar{R} |R(x) - \bar{R}| + |R(x) - \bar{R}|^2\}, \end{aligned}$$

where we have used that $-2D \leq \chi(h_{xy}) - D \leq 0$. Again by Hölder's inequality ($x \in P$ arbitrary)

$$\sup_{0 \leq t \leq 1} \|e^{tS_P}\|_{2r} \leq \|e^{2T\bar{R}|R(x)-\bar{R}|}\|_1^{1/4r} \cdot \|e^{T|R(x)-\bar{R}|^2}\|_1^{1/4r} \quad (4.23)$$

with $T = 16r\lambda D/(d-1) = (32dD) \cdot \lambda$. Inequalities (A6) and (A7) of the Appendix show that both factors on the right-hand side of (4.23) are bounded by constants just depending on α and the constants of the theory (d, D, k) but not on $\bar{R}$. This proves the first part, (4.21), of Lemma 4.3. We now prove (4.22). From (4.16) we get for xy any bond in P

$$\begin{aligned} \|S_P\|_{2r} &\leq (1/2g_0^2) \|\chi(h_P) - D\|_{2r} \\ &\quad + (2\lambda/(d-1)) \{\|\chi(h_P) - D\|_{4r} \|R(x) R(y) - \bar{R}^2\|_{4r} \\ &\quad + (D/2) \|R(x) - R(y)^2\|_{4r}\}. \end{aligned}$$

The terms involving $R(x)$ are dealt with by the methods of the Appendix.

$$\|R(x) R(y) - \bar{R}^2\|_{4r} \leq 2\bar{R} \|R(x) - \bar{R}\|_{4r} + \|R(x) - \bar{R}\|_{8r}^2 \quad (4.24)$$

and the right-hand side of (4.24) is bounded uniformly in $\bar{R}$ by (A8) and (4.20):

$$\begin{aligned} \|(R(x) - R(y))^2\|_{4r} &\leq \|R(x) - R(y)\|_{8r}^2 \\ &\leq 4 \|R(x) - \bar{R}\|_{8r}^2 \\ &\leq O(\bar{R}^{-1}), \quad \text{for large } \bar{R} \end{aligned}$$

by (A8) and (4.20).

To conclude the proof of (4.22) we need the following lemma.

LEMMA 4.4. *For all q , uniformly in c_1 and c_2*

$$\lim_{\bar{R} \rightarrow \infty} \|\chi(h_P) - D\|_q = 0.$$

Remark. One can actually show that $\|\chi(h_P) - D\|_q = O(\bar{R}^{-n})$ with n depending on the gauge group $\mathcal{G}$.

Proof. Let $U(\cdot)$ be the finite-dimensional unitary representation of $\mathcal{G}$ of which $\chi(\cdot)$

is the character, i.e., $\chi(g) = \text{tr } U(g)$. We now write $\chi(h + g)$ for $\text{tr}(U(h) + U(g))$, etc. For $h, g \in \mathcal{G}$ and e the identity element in $\mathcal{G}$ we find

$$\begin{aligned} |\chi(e - gh)| &= |\chi(e - g) + \chi(e - h) - \chi((e - g)(e - h))| \\ &\leq |\chi(e - g)| + |\chi(e - h)| + [\chi((e - g)(e - g)^*) \chi((e - h)(e - h)^*)]^{1/2}. \end{aligned}$$

On the other hand $\chi((e - g)(e - g)^*) = \chi(e - g) + \chi((e - g)^*)$ because $\chi(e - gg^*) = \text{tr}(1 - U(g)U(g^*)) = 0$. Hence

$$\begin{aligned} |\chi(e - gh)| &\leq |\chi(e - g)| + |\chi(e - h)| + 2[|\chi(e - g)| |\chi(e - h)|]^{1/2} \\ &\leq 2(|\chi(e - g)| + |\chi(e - h)|). \end{aligned}$$

By the same arguments we find for $h_p = h_{xy}h_{yu}h_{uv}h_{vx}$

$$\begin{aligned} |\chi(h_p) - D| &= |\chi(e - h_{xy}h_{yu}h_{uv}h_{vx})| \\ &\leq 4(|\chi(e - h_{xy})| + |\chi(e - h_{yu})| + |\chi(e - h_{uv})| + |\chi(e - h_{vx})|). \end{aligned}$$

This implies that

$$\|\chi(h_p) - D\|_q \leq 4 \|\chi(h_{xy}) - D\|_q. \quad (4.25)$$

From (4.10) to (4.13) we find (assuming χ to be real)

$$\|\chi(h_{xy}) - D\|_q^q = \frac{\int |\chi(g) - D|^q e^{\lambda \bar{R}^2(\chi(g) - D)} dg}{\int e^{\lambda \bar{R}^2(\chi(g) - D)} dg}.$$

Writing μ for $\lambda \bar{R}^2$ and ρ for $D - \chi(g)$, we get for any $\epsilon > 0$

$$\begin{aligned} \|\chi(h_{xy}) - D\|_q^q &= \frac{\int \rho^q e^{-\mu \rho} dg}{\int e^{-\mu \rho} dg} \\ &\leq \frac{\int_{\rho \leq (\epsilon/2)^{1/q}} \rho^q e^{-\mu \rho} dg}{\int e^{-\mu \rho} dg} + \frac{(2D)^q \int_{\rho \leq (\epsilon/2)^{1/q}} e^{-\mu \rho} dg}{\int_{\rho \leq \frac{1}{2}(\epsilon/2)^{1/q}} e^{-\mu \rho} dg} \\ &\leq \frac{\epsilon}{2} + (2D)^q e^{-\frac{1}{2}\mu(\epsilon/2)^{1/q}} \left(\int_{\rho \leq \frac{1}{2}(\epsilon/2)^{1/q}} dg \right)^{-1} \\ &\leq \epsilon \quad \text{for } \mu \text{ sufficiently large.} \end{aligned}$$

This together with (4.25) proves Lemma 4.4.

5. WILSON'S CONFINEMENT BOUND

It has become one of the most challenging problems of elementary particle physics to explain why the quarks, though the elementary constituents of the material world,

can never be observed experimentally. It is generally believed (or hoped) that non-abelian gauge theories do “confine” quarks, but there are still rather widespread views on why this should happen [4, 18, 28, 35]. Wilson [35] claims that confinement follows from a bound of the form

$$\langle \chi_\sigma(C) \rangle \leq e^{-O(\text{area } C)}, \quad (5.1)$$

where C is a closed path on the lattice (e.g., a rectangle), $\text{area } C$ is the area of the surface spanned by C , and χ_σ is some character on the gauge group $\mathcal{G}$. (Remember: if $C = (x_1 x_2 x_3 \cdots x_n = x_1)$, then $\chi_\sigma(C) = \chi_\sigma(g_{x_1 x_2} g_{x_2 x_3} \cdots g_{x_{n-1} x_1})$). For simplicity we will always assume that C is planar and does not intersect itself, e.g., $C = \text{rectangle}$. Then $\text{area } C$ is simply the number of plaquettes enclosed by C . In this section we want to show that in the region of convergence of the cluster expansion (5.1) holds. We limit ourselves to theories with gauge group $\mathcal{G} = SU(n)$ or $\mathcal{G} = U(1)$.

Let D_σ be an irreducible representation of $SU(n)$ with Young tableau Y_σ . We define

$$n_\sigma = (1/n) \{ \text{number of boxes in } Y_\sigma, \text{ mod } n \}.$$

(For $SU(3)$, n_σ is known as triality.) For $\mathcal{G} = U(1)$, we define $n_\sigma = \sigma$, where $D_\sigma(e^{i\varphi}) = e^{i\sigma\varphi}$, $\sigma \in \mathbb{Z}$. Also, $\chi_\sigma(g) = \text{tr } D_\sigma(g)$.

THEOREM 5.1. *In the region of convergence of the cluster expansion in a pure Yang–Mills theory, (5.1) holds provided that $n_\sigma \neq 0$.*

Proof. The strategy is to show that the cluster expansion of the left-hand side of (5.1) contains only terms in which $|Q| \geq \text{area } C$. Then by Theorem 3.1, a factor $\epsilon^{\text{area } C}$, $\epsilon < 1$, can be extracted from each term in the cluster expansion. (This is analogous to the proof of the existence of a mass gap.) All we have to prove is the following technical lemma.

LEMMA 5.2. *Let ρ_P be any invariant L^∞ function of the plaquette variable g_P and assume $n_\sigma \neq 0$. Then*

$$\int \chi_\sigma \left(\prod_{xy \in C} g_{xy} \right) \prod_{P \in Q} \rho_P \prod dg_{xy} = 0$$

for all Q with $|Q| < \text{area } C$.

For a proof we use the Peter–Weyl expansion (Fourier expansion) of ρ_P (see, e.g., [17, 27, 29]):

$$\rho_P(g_P) = \sum_{\sigma} a_{\sigma} \chi_{\sigma}(g_P).$$

Therefore Lemma 5.2 follows if we can show that

$$\int \chi_\sigma \left(\prod_{xy \in C} g_{xy} \right) \prod_{P \in Q} \chi_{\tau_P}(g_P) \prod dg_{xy} = 0 \quad (5.2)$$

for $|Q| < \text{area } C$ and any choice of τ_P , $P \in Q$. Concentrating in (5.2) on the integration over the variable g_{xy} of a fixed bond xy we find (again by Peter-Weyl) that we get zero unless

$$\sum_{P: xy \in P} \epsilon_{P,xy} n_{\tau_P} = 0 \quad \text{for } xy \notin C, \quad (5.3)$$

$$\sum_{P: xy \in P} \epsilon_{P,xy} n_{\tau_P} = \pm n_\sigma \quad \text{for } xy \in C. \quad (5.4)$$

Here $\epsilon_{P,xy} = \pm 1$, depending on whether g_{xy}^{-1} or g_{xy} is a factor in g_P . Equations (5.3)–(5.4) are to be understood as congruences mod 1. Let us first assume that Q lies entirely in the plane of C . Then each bond xy belongs to at most two plaquettes P' and P'' in Q and $\epsilon_{P',xy} = -\epsilon_{P'',xy}$. Hence (5.3) reduces to $n_{\tau_{P'}} = n_{\tau_{P''}}$ if both P' and P'' belong to Q and $n_{\tau_{P'}} = 0$ if P' contains at least one bond not contained in any $P'' \in Q$. This shows that $n_{\tau_P} = 0$ for P outside C and $n_{\tau_P} = \text{constant}$ for P in Q and inside C . (5.4) now forces this constant to be $\pm n_\sigma \neq 0$. This leads to a contradiction unless all the plaquettes inside C belong to Q , hence (5.2). Now let us look at a more general set Q .

We say that two bonds xy and $x'y'$ are equivalent, $xy \sim x'y'$, if they are parallel and if both lines xx' and yy' are perpendicular to the plane of the loop. We now pick a bond xy in the plane of C and add Eqs. (5.3)–(5.4) for all $x'y' \sim xy$. The contributions from plaquettes P which are orthogonal to the plane of C drop out from the sums as they contain exactly two bonds $x'y'$ and $x''y''$ equivalent to xy , but $\epsilon_{P,x'y'} = \epsilon_{P,x''y''}$. Hence we are left with

$$\sum_{\substack{P: x'y' \in P \\ x'y' \sim xy \\ P \parallel \text{plane of } C}} \epsilon_{P,x'y'} n_{\tau_P} = 0 \quad \text{for } xy \notin C,$$

$$\sum_{\substack{P: x'y' \in P \\ x'y' \sim xy \\ P \parallel \text{plane of } C}} \epsilon_{P,x'y'} n_{\tau_P} = \pm n_\sigma \quad \text{for } xy \in C.$$

Now the argument goes as before. This completes the proofs of Lemma 5.2 and Theorem 5.1.

Remark. For $\mathcal{G} = U(1)$ the bound on $\langle \chi_\sigma(C) \rangle$ has been studied in [11] for weak coupling and in [6] for strong coupling.

6. DISCUSSION

So far we have treated various gauge field models on a fixed lattice. Of course the real interest lies in the continuum limit.

One would like to let the lattice spacing tend to zero while keeping the masses of physical objects (“mesons,” “baryons”) fixed. By a simple scaling argument

$$m_i = a^{-1} f_i(g_0), \quad (6.1)$$

where g_0 stands for the bare coupling constant(s) of the theory and a is the lattice spacing. So when $a \rightarrow 0$ we need $f_i(g) \rightarrow 0$; that means if we were to look at the theory with a microscope that brings the lattice spacing back to 1, we would see the theory approaching a critical point, as has been noticed by various people; see, e.g., [12, 30].

If the map $\{g_0\} \mapsto \{m_i\}$ is invertible, it can be used to adjust the bare coupling(s) in such a way that a finite limit for the physical masses results; this coupling constant renormalization is the only kind of renormalization consistent with gauge invariance. A pure gauge theory has only one coupling constant; the "renormalization map" is therefore much simpler in this case than in, say, the φ^4 theory where it is a map from a portion of $\mathbb{R}^3$ into $\mathbb{R}^3$ (see [30]). This fact and, of course, asymptotic freedom, should make the continuum limit easier in gauge theories.

Of course, so far we know little about $f_i(g)$, except in the area where we have convergent expansions. In a pure Yang-Mills theory in low dimension ($d \leq 4$) we expect a qualitative behavior of the type illustrated in Fig. 2 where $\lim_{g_0 \rightarrow \infty} f(g_0) = \infty$, $\lim_{g_0 \rightarrow 0} f(g_0) = 0$, $f(g_0) \sim 4 \ln g_0$ for large g_0 . In the case of an abelian group ($U(1)$) there may be a critical point at a finite coupling constant g_c in $d = 4$ (see Fig. 3). The popular belief is that in fact for $g_0 \leq g_c$, the theory is "critical" in the sense that there is no mass gap. In more than four dimensions a critical point at finite

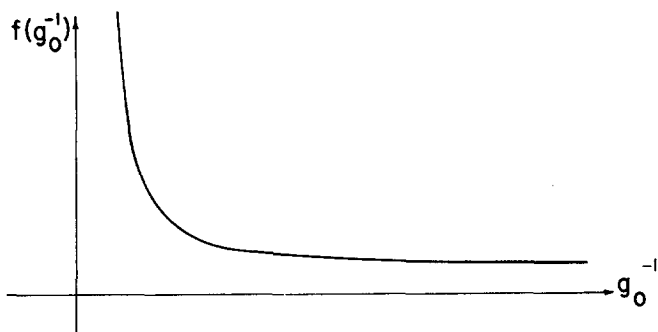


FIGURE 2

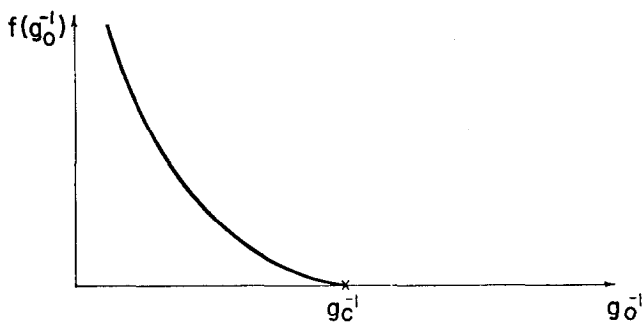


FIGURE 3

g_0^{-1} should occur also in the non-abelian case. It should be noted that these critical points (if they exist) do not indicate that there is spontaneous breakdown of gauge invariance for $g_0^{-1} > g_c^{-1}$; in that respect the situation is quite different from the one encountered in the more usual lattice systems [9]. The precise statement is the following: In an equilibrium state fulfilling the DLR equations [19], there is no spontaneous breakdown of gauge invariance. This is almost trivial since the DLR equations state that equilibrium states restricted to a finite volume are identical to Gibbs states with certain boundary conditions. Since these finite volume Gibbs states are still gauge invariant (under gauge transformations that affect the interior of the region only) they cannot produce gauge-dependent expectations for observables localized inside. See also [22].

The argument just presented works of course only in the gauge-invariant formalism. If we “fix a gauge” (e.g., use the radiation gauge), the situation is different. This explains why in a continuum theory (for instance, free QED) gauge invariance is always broken (see [33]).

In order to get some idea of what should be expected for weak coupling or small lattice spacing, it is instructive to look at a simplified Yang–Mills model that is exactly soluble and is exact in two dimensions. To define the model we drop from the action (2.2) all “horizontal” plaquettes, that is, all plaquettes not containing any bond in “time” direction. The transfer matrix T then simplifies to a tensor product of copies of an operator t ; $t^{1/2}$ is given by the kernel (in the radiation gauge)

$$t^{1/2}(h, g) = \exp((1/2g_0^2) \operatorname{Re} \chi_r(hg^{-1}) + \alpha), \quad (6.2)$$

where α is chosen in such a way that the largest eigenvalue of t is 1. $t^{1/2}$ (and therefore t) is easily diagonalized by applying the Peter–Weyl theorem [17, 27, 29]:

$$t^{1/2}(h, g) = \sum_{\sigma} \lambda_{\sigma}(g_0) \chi_{\sigma}(hg^{-1}), \quad (6.3)$$

$$\lambda_{\sigma}(g_0) = \int t^{1/2}(h, e) \bar{\chi}_{\sigma}(h) dh, \quad (6.4)$$

where $\{\chi_{\sigma}\}$ is a complete set of characters of the group $\mathcal{G}$. The eigenvalues of $t^{1/2}$ are just

$$\hat{\lambda}_{\sigma} = \lambda_{\sigma}/d_{\sigma}, \quad (6.5)$$

where $d_{\sigma} = \chi_{\sigma}(e)$ is the dimension of the representation D_{σ} . All eigenvalues are nonnegative (this follows from the general discussion in Section 2 but more simply from the fact that $\hat{t}$ is of positive type. We order the eigenvalues

$$1 = \hat{\lambda}_{\sigma_0} \geq \hat{\lambda}_{\sigma_1} \geq \dots. \quad (6.6)$$

(By a Perron–Frobenius argument it can be seen that σ_0 is the trivial representation.)

If we interpret $T^{1/2}$ as e^{-aH} on $\mathcal{H}$ we find that H has eigenvalues of the form

$$\epsilon(\{n_\sigma\}) = -\frac{1}{a} \sum_\sigma n_\sigma \ln \hat{\lambda}_\sigma, \quad (6.7)$$

where n_σ are nonnegative integers. The mass gap on the Hilbert space $\mathcal{H}$ is therefore

$$m = -(1/a) \ln \hat{\lambda}_\sigma. \quad (6.8)$$

On the gauge-invariant subspace ee_i the mass gap obeys, as can be seen easily

$$m_{g,i} \geq -(4/a) \ln \hat{\lambda}_\sigma. \quad (6.9)$$

(In two dimensions actually

$$m_{g,i} = +\infty \quad (d=2), \quad (6.10)$$

expressing that nonoverlapping gauge-invariant observables are independent.)

The energy of a “string” of length L belonging to the representation τ is

$$\epsilon_L = -(L/a^2) \ln \hat{\lambda}_\tau. \quad (6.11)$$

We would like to have a finite limit for quantities like (6.11) as $a \rightarrow 0$. That forces $g_0 \rightarrow 0$ (see (6.4), (6.2)). A Gaussian approximation for

$$\lambda_\sigma(g_0) = \int \exp\left(-\frac{1}{2g_0^2} \chi_\tau(h)\right) \chi_\sigma(h) dh / \int \exp\left(-\frac{1}{2g_0^2} \chi_\tau(h)\right) dh \quad (6.12)$$

gives the correct behavior for $\lambda_\sigma(g_0)$ for small g_0 ; the result is

$$\lambda_0(g_0) = d_\sigma - c g_0^2 + O(g_0^3) \quad (6.13)$$

and therefore

$$\epsilon_L = -\frac{L}{a^2} \ln\left(1 - \frac{c}{d_\sigma} g_0^2 + O(g_0^3)\right) = \frac{L}{a^2} \frac{c}{d_\sigma} g_0^2 + O(g_0^3) \quad (6.14)$$

so if we choose g_0 proportional to a we obtain a finite limit for ϵ_L/L as $a \rightarrow 0$.

The same prescription $g_0 \sim a$ also produces a finite limit for the expectations of loop variables. Consider $d=2$ for simplicity. If g_L is a closed-loop variable we obtain

$$\langle \chi_\tau(g_L) \rangle = (\hat{\lambda}_\tau)^{a^{-2} \cdot \text{area } L}, \quad (6.15)$$

where area L is the area enclosed by the loop L . By choosing $g_0 \sim a$ this becomes in the continuum limit (P meaning “path ordering”)

$$\left\langle \chi_\tau \left(P \exp \left(i \oint_L A^\mu dx_\mu \right) \right) \right\rangle = C_\tau^{\text{area } L} \quad (6.16)$$

with some constant $C_\tau < 1$. This agrees with the result that can be computed directly in the case of $\mathcal{G} = U(1)$ by Gaussian integration.

As remarked before, loop variables belonging to nonoverlapping loops are statistically independent. So (6.16) can be considered a complete description of the two-dimensional continuum Yang-Mills theory, expressed in terms of the somewhat unusual loop variable. Only in the abelian case is it easy to recover the Euclidean quantum field theory from these loop variables since

$$\exp \left(i \oint_{\partial G} A^\mu dx_\mu \right) = \exp \left(i \oint_G F d^2x \right), \quad (6.17)$$

$$F = \epsilon^{\mu\nu} F_{\mu\nu}.$$

F of course is a trivial white noise field; its corresponding relativistic field vanishes as has to be the case. The non-Abelian (Euclidean) Yang-Mills field cannot be reconstructed from the loop variables in such a way; but it is possible that these loop variables are the more natural objects in a Yang-Mills theory. It is an interesting question whether it is possible to find a consistent field theoretic framework based on these "loops" instead of local fields.

APPENDIX

We prove several elementary estimates which are necessary to establish Lemma 4.3. The main input will be assumptions (4.2) on the potential V and conditions (4.20) on the constants c_1 and c_2 . We will always assume that $\bar{R} > 1$. By γ we denote a constant that depends on k only (its exact value might change from line to line). We assume $k \geq 1$.

$$\begin{aligned} \text{(a)} \quad \int_0^\infty e^{-V(R)} R^k dR &\geq \int_{\bar{R}/2}^{3\bar{R}/2} e^{-c_2(x-\bar{R})^2} x^k dx \\ &\geq \gamma \bar{R}^k \int_0^{\bar{R}/2} e^{-c_2 x^2} dx \\ &\geq \gamma \bar{R}^k c_2^{-1/2}. \end{aligned} \quad (\text{A1})$$

$$\begin{aligned} \text{(b)} \quad R^k &\leq (2\bar{R})^k + (2|R - \bar{R}|)^k \\ &\leq \gamma \bar{R}^k e^{|R - \bar{R}|^2}. \end{aligned} \quad (\text{A2})$$

$$\begin{aligned} \text{(c)} \quad \int_0^\infty e^{\beta|R - \bar{R}| - V(R)} R^k dR &\leq \gamma \bar{R}^k \int_0^\infty e^{\beta|R - \bar{R}| - (c_1/2)(R - \bar{R})^2} dR \\ &\leq \gamma \bar{R}^k e^{(\beta^2/2c_1)} c_1^{-1/2}. \end{aligned} \quad (\text{A3})$$

$$\begin{aligned}
 \text{(d)} \quad \int_0^\infty e^{\beta|R-\bar{R}|^2 - V(R)} R^k dR &\leq \gamma \bar{R}^k \int_0^\infty e^{-((c_1/2) - \beta)|R-\bar{R}|^2} dR \\
 &\leq \gamma \bar{R}^k (c_1 - 2\beta)^{-1/2},
 \end{aligned} \tag{A4}$$

assuming $2\beta < c_1$.

$$\begin{aligned}
 \text{(e)} \quad \int_0^\infty |R - \bar{R}|^q e^{-V(R)} R^k dR &\leq \gamma \bar{R}^k \int_{-\infty}^\infty |R - \bar{R}|^q e^{-(c_1/2)(R-\bar{R})^2} dR \\
 &\leq \gamma \bar{R}^k c_1^{-(q+1)/2}
 \end{aligned} \tag{A5}$$

(with γ depending on k and q). Now we combine these estimates to prove Lemma 4.3. From (A1) and (A3)–(A5) we get

$$\|e^{2T\bar{R}|R(x)-\bar{R}|}\|_1^{1/4r} \leq \gamma e^{(T\bar{R})^2/2rc_1} (c_2/c_1)^{1/8r} \tag{A6}$$

$$\|e^{T|R(x)-\bar{R}|^2}\|_1^{1/4r} \leq \gamma (c_2/(c_1 - 2T))^{1/8r} \tag{A7}$$

$$\|R(x) - \bar{R}\|_{8r} \leq \gamma (c_2/c_1)^{1/16r} c_1^{-1/2} \tag{A8}$$

ACKNOWLEDGMENT

It is a pleasure to thank Professor W. Zimmermann for the hospitality at the Max-Planck-Institut für Physik und Astrophysik (München), where part of this work has been carried out.

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